

Question

The nitrate of metal **M** reacts with 1,4-benzenedicarboxylic acid (abbreviated as H₂BDC) in a mixed system of N,N-dimethylformamide (abbreviated as DMF) and anhydrous ethanol to yield a valuable material **A**. Single-crystal X-ray diffraction analysis of **A** reveals that **A** belongs to the orthorhombic crystal system, with unit cell parameters $a = 1923.01 \text{ pm}$, $b = 723.87 \text{ pm}$, $c = 884.82 \text{ pm}$, $Z = 4$, and a density of $\rho = 1.597 \text{ g/cm}^3$.

The framework structure of **A** primarily consists of **M** and BDC²⁻, with solvent molecules incorporated in the structure. Elemental analysis indicates that **A** contains five elements: **M**, C, H, O, and N, with an atomic ratio of **M** to C of 1:11. When **A** is subjected to high-temperature oxidation in air, nanoparticles of a common oxide of **M** are obtained, and the mass fraction of **M** in this common oxide is approximately 70%.

Based on the above information, determine **M**.

A. Co

B. Ni

C. Cu

D. Ca

E. Fe

F. Na

Answer

Correct Answer: **A**

Detailed Explanation

Based on the crystallographic data of material **A**, the chemical formula of **A** can be inferred. From $Mr = \frac{\rho N_{AV}}{Z}$, substituting the data yields a molecular weight of 296.03 g/mol for this material.

CHECKPOINT

1 PTS

$$Mr(\mathbf{A}) = 296.03 \text{ g/mol}$$

The problem states that the ratio of **M** to **C** in **A** is 1:11, which can be used to deduce the chemical composition of **A**. Assuming for now that the chemical formula of **A** contains only one **M**, this would correspond to 11 **C** atoms. One $(\text{BDC})^{2-}$ contains 8 **C** atoms, so the formula should include only one $(\text{BDC})^{2-}$. This leaves 3 **C** atoms unaccounted for, which matches the carbon count of one DMF molecule. If the chemical formula is $\text{M}(\text{BDC})(\text{DMF})$, calculating the molecular weight gives a metal molecular weight of 58.80 g/mol.

CHECKPOINT

1 PTS

$$Mr(\mathbf{M}) = 58.80 \text{ g/mol}$$

According to the periodic table, the metals that meet this requirement could be Co or Ni. Based on the given conditions, when this metal is oxidized at high temperatures in air, it forms a common oxide with a mass fraction of the metal of approximately 70%. Possible oxides of Co include CoO , Co_2O_3 , and Co_3O_4 , while the oxide of Ni is NiO . Considering the mass fraction, the metal is Co.

CHECKPOINT

1 PTS

Based on the mass fraction, the metal is Co